Math 521, Section 2
Homework 11

Problem 1. Consider the metric space $(\mathbb{R}, d)$ with the usual metric $d(x, y) = |x - y|$. Note that the set $\mathbb{Q}$ is not an interval, and so from lecture we know that it must be disconnected.

(a) Find two sets $A, B \subseteq \mathbb{R}$ that are nonempty and separated so that $\mathbb{Q} = A \cup B$, and prove your answer.
(b) Find a set $\emptyset \neq C \subsetneq \mathbb{Q}$ that is both open and closed in $\mathbb{Q}$, and prove your answer.

Problem 2. Let (X, d) be a metric space.

(a) Prove that if $F \subseteq X$ is disconnected and closed in X, then there exist $A, B \subseteq X$ that are nonempty, separated, and closed in X so that $F = A \cup B$.
(b) Prove that if $U \subseteq X$ is disconnected and open in X, then there exist $A, B \subseteq X$ that are nonempty, separated, and open in X so that $U = A \cup B$.

Problem 3. Suppose that (X, d) is a metric space such that for any pair of points $a, b \in X$, there exists a connected set $C \subseteq X$ so that $a, b \in C$. Prove that X is connected.

Problem 4. Let (X, d) be a metric space and $A \subseteq X$. Prove that if A is connected, then $\overline{A}$ is connected.

Problem 5. Let (X, d) be a metric space and $A, B \subseteq X$. Prove that if A and B are both connected and $A \cap B \neq \emptyset$, then $A \cup B$ is connected.

Problem 6. Let (X, d) be a metric space.

(a) Let $A, B, C \subseteq X$. Prove that if the sets A and B are separated and the sets A and C are separated, then the sets A and $B \cup C$ are separated.
(b) Prove that if $F, G \subseteq X$ are closed and $F \cup G$ and $F \cap G$ are connected, then F is connected.